DESCENT OF FINITE FREENESS

Abstract. Let M be an R-module whose underlying type is an étale sheaf. We prove that if M is étale-locally free of a fixed finite rank, then it merely has a basis of that rank. We also prove descent of a basis together with a linear retraction from a finite free ambient module.

1. Setting and statement

We work in synthetic algebraic geometry over R, with locality, synthetic quasi-coherence, and Zariski-local choice as in [1]. In particular, R is a commutative local ring, $1 \neq 0$, and for every finitely presented commutative R-algebra A, evaluation is an isomorphism

$$A \longrightarrow (\operatorname{Spec} A \longrightarrow R).$$

Here $\operatorname{Spec} A$ is the type of R-algebra homomorphisms $A \to R$.

All existence assertions are understood constructively: $\exists z$ asserts that a witness merely exists. The notation $\|Z\|$ likewise asserts that Z is inhabited, without choosing an element of Z. An isomorphism of modules always means an R-linear isomorphism. The underlying type of a module is a set, so its equalities are propositions.

For a monic polynomial $f \in R[t]$ of degree $d > 0$, put

$$B_f = R[t]/(f), \qquad X_f = \operatorname{Spec} B_f = \{x \in R : f(x) = 0\}, \qquad G_f = \|X_f\|.$$

The topology T is generated by the propositions G_f. A type Y is a *sheaf* if the constant map

$$Y \longrightarrow (G_f \longrightarrow Y)$$

is an equivalence for every f. For a proposition P, this says precisely that $(G_f \to P) \to P$ for every f.

We assume propositional sheafification: $\|Z\|_T$ is a proposition and a sheaf, receives a map from Z, and any map from Z to a sheaf proposition factors through $\|Z\|_T$. Thus a sheaf proposition may be proved under a hypothesis which holds T-locally.

Write $\mathcal{B}_r(M)$ for the type of ordered bases $(v_1, \dots, v_r)$ of M, or equivalently of linear isomorphisms $R^r \to M$.

Theorem 1.1. *Let $r \geq 0$ be fixed and let M be an R-module whose underlying type is a sheaf. Then*

$$\|\mathcal{B}_r(M)\|_T \longrightarrow \|\mathcal{B}_r(M)\|.$$

We shall use the following results about schemes. They follow from the SAG axioms and do not involve descent of modules.

(1) Zariski-local choice on an affine $X = \operatorname{Spec} A$ gives finitely many $g_i \in A$ generating the unit ideal, together with actual choices on each $D(g_i)$, from a family of merely inhabited types on X. The same conclusion, with a finite cover by open affine charts, holds on a scheme.

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(2) If $P_n(x)$ is an increasing sequence of propositions on a scheme X, then

$$(\forall x \in X\ \exists n\ P_n(x)) \implies \exists N\ \forall x \in X\ P_N(x).$$

This is the uniform bound theorem for schemes. Intersections of open charts are schemes, so it applies to these intersections as well.

(3) An open predicate on an affine scheme is a finite union of standard opens. A finite family $D(g_i)$ covers $\operatorname{Spec} A$ exactly when the g_i generate 1 in A. Open propositions are stable under double negation.

The fraction argument below is the argument for sections in [1, Section 7.1]. We include it to specify exactly what is used.

Sheaves are closed under products, dependent products, dependent sums, and equality types. Consequently, if Y and Z are sheaves and $u : Y \to Z$ is fixed, then "u is an equivalence" is a sheaf proposition. Indeed, express it by a left inverse and a right inverse with their equations; the function types and equality types in this description are sheaves.

The ring R is a sheaf. Here is the coefficient argument. The algebra B_f is free on $1, t, \dots, t^{d-1}$, and evaluation identifies it with R^{X_f}. If $h : G_f \to R$, let $b \in B_f$ represent $x \mapsto h(|x|)$. For each $x \in X_f$, the element b is equal to $h(|x|) \cdot 1$: evaluate at any $y \in X_f$ and use that G_f is a proposition. The constant coefficient of b is therefore a value of h. Uniqueness follows from injectivity of $R \to B_f$ and of evaluation.

Lemma 1.2. *Let Y be a set which is a sheaf, and put $X = X_f$. If $u : X \to Y$ satisfies $u(x) = u(y)$ for all $x, y \in X$, there is a unique $v \in Y$ with $v = u(x)$ for every $x \in X$. Also, an equality between elements of a sheaf follows if it holds for every $x \in X$.*

Proof. Given $q : \|X\|$, the common value of u exists and is unique, so it can be chosen uniquely. This gives a map $\|X\| \to Y$, which has a unique constant value because Y is a sheaf. For the second assertion use that equality types of a sheaf are sheaves. □

2. Fractions of module-valued functions

For $a \in R$, the notation $D(a)$ means the proposition that a is a unit. For a function $g : X \to R$, it means the subtype of X where $g(x)$ is a unit.

Definition 2.1. An R-module L satisfies the *fraction condition*, denoted $\mathcal{F}(L)$, if the following hold for every $a \in R$.

(1) For every $s : D(a) \to L$, there exist $n \geq 0$ and $u \in L$ such that $s = u/a^n$ on $D(a)$.
(2) If $u \in L$ is zero on $D(a)$, there exists $n \geq 0$ such that $a^n u = 0$.

Both clauses assert mere existence. In the terminology of [1, Definition 7.1.5], this is weak quasi-coherence.

The condition $\mathcal{F}(L)$ is a proposition. It is preserved by linear isomorphisms and finite products. For products, choose finitely many exponents and raise them to a common exponent. The module R satisfies it: use evaluation for $R[1/a]$, together with the elementary fraction and kernel descriptions of this localization. Hence every finite free module satisfies it.

Lemma 2.2 (Fractions of sections)**.** *Let X be a scheme and let L_x be an R-module for each $x \in X$, with $\mathcal{F}(L_x)$ for every x. For $g : X \to R$, the following hold.*

(1) *Every section s of L_x over $D(g)$ can be written*

$$s(x) = u(x)/g(x)^n$$

for some $n \geq 0$ and some section u on all of X.

(2) *If a section u on X vanishes on $D(g)$, then $g^n u = 0$ on X for some $n \geq 0$.*

In particular, the module $\prod_{x \in X} L_x$ satisfies the fraction condition over R.

Proof. At each x, the first fraction clause gives a numerator and an exponent for $s(x)$. The assertion that an exponent at most n suffices is increasing in n. The uniform bound theorem therefore gives one exponent n which suffices everywhere. Zariski-local choice then gives a finite open cover U_i of X and numerators u_i on U_i, with $u_i = g^n s$ on $U_i \cap D(g)$.

On $U_i \cap U_j$, the difference $u_i - u_j$ vanishes on $D(g)$. The second fraction clause, followed by the uniform bound theorem on this intersection, gives an exponent which kills that difference everywhere on the intersection. Choose a common exponent k for the finitely many pairs. The sections $g^k u_i$ now agree on overlaps, and hence give a section u on X. On $D(g)$ it satisfies $u = g^{n+k} s$.

For the second assertion apply the second fraction clause at every x, and then apply the uniform bound theorem to $g(x)^n u(x) = 0$. Taking g constant and exchanging the section arguments proves the last assertion. □

Lemma 2.3 (Kernels)**.** *If P and Q satisfy the fraction condition and $u : P \to Q$ is linear, then* $\ker u$ *satisfies the fraction condition.*

Proof. Let $s : D(a) \to \ker u$. In P, write $s = p/a^n$. The element $u(p)$ vanishes on $D(a)$, so $a^k u(p) = 0$ for some k. Thus $a^k p \in \ker u$ and $s = (a^k p)/a^{n+k}$. The second fraction clause is inherited from P. □

Proposition 2.4. *If the underlying type of M is a sheaf, then the proposition $\mathcal{F}(M)$ is a sheaf.*

Proof. Fix f and put $X = X_f$. Suppose $G_f \to \mathcal{F}(M)$. In particular, for each $x \in X$ we have $\mathcal{F}(M)$. Lemma 2.2 gives the fraction condition for

$$P = M^X, \qquad Q = P^X = M^{X \times X}.$$

Here we apply the lemma first to obtain the condition for P, and then again to obtain it for Q. We prove the two clauses for M.

Let $a \in R$ and $s : D(a) \to M$. Apply the first clause for P to the function which is constant in x. There exist $n \geq 0$ and $b : X \to M$ such that

$$s = b(x)/a^n \qquad \text{on } D(a), \quad x \in X.$$

Hence $b(x) - b(y)$ vanishes on $D(a)$. Apply the second clause for Q to this function of (x, y). For some $k \geq 0$,

$$a^k(b(x) - b(y)) = 0 \qquad (x, y \in X).$$

The function $x \mapsto a^k b(x)$ is constant, so Lemma 1.2 supplies $m \in M$ with $m = a^k b(x)$ for all x. On $D(a)$, the equality

$$s = m/a^{n+k}$$

holds for every $x \in X$. Since equality in M is a sheaf, it holds without the hypothesis $x \in X$.

For the second clause, suppose $m \in M$ vanishes on $D(a)$. Apply the second clause for P to the constant function with value m. It gives $a^n m = 0$ for every $x \in X$, with one exponent n. Descend this equality by Lemma 1.2. □

Corollary 2.5. *If M is a sheaf and $\|\mathcal{B}_r(M)\|_T$ holds, then $\mathcal{F}(M)$ holds. Thus Lemma 2.2 applies to the constant family M on every scheme.*

Proof. The proposition $\mathcal{F}(M)$ is a sheaf, so we may prove it locally, where $M \cong R^r$. □

3. Choosing solutions of linear equations

The following affine gluing result is the additive statement of [1, Theorem 7.3.6]. We give its fraction proof.

Lemma 3.1. *Let $X = \operatorname{Spec} A$ be affine and let K_x be modules satisfying the fraction condition. Suppose E_x is a merely inhabited set on which addition by K_x is defined, with*

$$e + 0 = e, \qquad (e + k) + l = e + (k + l),$$

and, for every $e, e' \in E_x$, exactly one $k \in K_x$ satisfies $e + k = e'$. Then there merely exists a section $x \mapsto e(x) \in E_x$.

Proof. By Zariski-local choice, choose $g_1, \dots, g_p \in A$ generating 1, and sections e_i on $D(g_i)$. On an overlap write

$$\delta_{ij} = e_i - e_j \in K_x.$$

These differences satisfy $\delta_{ij} + \delta_{jk} = \delta_{ik}$ on triple overlaps. By Lemma 2.2, and finite choice of a common exponent,

$$\delta_{ij} = \frac{b_{ij}}{(g_i g_j)^n} \quad \text{on } D(g_i g_j)$$

for sections b_{ij} of K_x on X and one $n \geq 0$. The sections

$$g_i^n b_{jk} - g_j^n b_{ik} + g_k^n b_{ij}$$

vanish on $D(g_i g_j g_k)$. Apply the second clause of Lemma 2.2, and choose a common exponent q for all triples. Put

$$F_i = g_i^{n+q}, \qquad \mu_{ij} = (g_i g_j)^q b_{ij}.$$

We then have, on all of X,

$$F_i \mu_{jk} - F_j \mu_{ik} + F_k \mu_{ij} = 0. \tag{1}$$

The elements F_i still generate 1. Indeed, powers of finitely many elements generating 1 again generate 1, by expanding a sufficiently high power of an identity $\sum_i a_i g_i = 1$. Choose $\rho_i \in A$ with $\sum_i \rho_i F_i = 1$, and set

$$V_i = \sum_k \rho_k \mu_{ik}.$$

Equation (1) gives

$$F_j V_i - F_i V_j = \sum_k \rho_k F_k \mu_{ij} = \mu_{ij}.$$

On $D(g_i)$ let $t_i = V_i / F_i$. On an overlap,

$$t_i - t_j = \frac{\mu_{ij}}{F_i F_j} = \delta_{ij}.$$

Consequently the sections $e_i - t_i$ agree on overlaps and give a section of E_x on X. □

Lemma 3.2. *Let $u : P \to Q$ be linear, with P and Q satisfying the fraction condition. Let X be affine and $q : X \to Q$. If*

$$\forall x \in X \; \exists p \in P \quad u(p) = q(x),$$

then there merely exists $p : X \to P$ with $u(p(x)) = q(x)$ for all x.

Proof. Put $K = \ker u$, which satisfies the fraction condition by Lemma 2.3. For each x, take

$$E_x = \{p \in P : u(p) = q(x)\}.$$

Addition by K preserves E_x, and the difference of two elements of E_x is the unique element of K taking one to the other. Apply Lemma 3.1. □

4. Coefficients and membership in a span

In this section M is a sheaf and we fix

$$\|\mathcal{B}_r(M)\|_T. \tag{2}$$

The assumptions remain in force until the end of the proof of Theorem 1.1.

Lemma 4.1. *For f monic of degree $d > 0$ and $X = X_f$, the map*

$$E_M : M^d \longrightarrow M^X, \qquad (m_0, \dots, m_{d-1}) \longmapsto \left(x \longmapsto \sum_{k=0}^{d-1} x^k m_k \right)$$

is a linear isomorphism.

Proof. The source and target are sheaves, so the assertion that this fixed map is an equivalence is a sheaf proposition. We may therefore use (2) and take $M = R^r$. In each coordinate the assertion is evaluation

$$R^d \cong R[t]/(f) \cong R^X.$$

Linearity is immediate from the formula. □

For $h : X \to M$, define $c_M(h) \in M$ to be the coefficient of x^0 in the unique expansion in Lemma 4.1. This map is linear and has the two properties

$$c_M(x \mapsto m) = m, \qquad c_M(x \mapsto a(x)m) = c_R(a)m. \tag{3}$$

For the first, the expansion is $(m, 0, \dots, 0)$. For the second, if $a(x) = \sum_k x^k a_k$, then $a(x)m = \sum_k x^k (a_k m)$.

Proposition 4.2. *For fixed $w_1, \dots, w_s, m \in M$, the proposition*

$$m \in \mathrm{Span}_R(w_1, \dots, w_s) \quad \Longleftrightarrow \quad \exists a_1, \dots, a_s \in R \quad m = \sum_i a_i w_i$$

is a sheaf.

Proof. Fix f and suppose this proposition holds under G_f. It then holds for each $x \in X_f$. Apply Lemma 3.2 on X_f to

$$R^s \longrightarrow M, \qquad (a_i) \longmapsto \sum_i a_i w_i,$$

and the constant function with value m. The source satisfies the fraction condition, and so does M by Corollary 2.5. We obtain functions $a_i : X_f \to R$ such that $m = \sum_i a_i(x) w_i$ for every x. Apply c_M to this equality and use (3):

$$m = \sum_i c_R(a_i) w_i.$$

This proves the required implication $(G_f \to P) \to P$. □

5. Selecting a basis

An open proposition is one which can be expressed as a finite disjunction of assertions that specified elements of R are units. Finite unions and finite intersections of open propositions are open.

Lemma 5.1. *For a matrix over R, surjectivity is an open proposition. For a square matrix, surjectivity is equivalent to invertibility. If $w_1, \dots, w_s$ generate R^r, then some r of them form a basis.*

Proof. For a matrix with columns $w_i = (w_i', a_i) \in R^{r-1} \times R$, a solution of $\sum_i \lambda_i w_i = (0, 1)$ gives $\sum_i \lambda_i a_i = 1$. By locality, some $a_j = u$ is a unit. Put

$$z_i = uw_i' - a_i w_j' \in R^{r-1}.$$

If the original matrix is surjective, the z_i generate R^{r-1}: lift $(u^{-1}y, 0)$ to obtain a combination of the z_i equal to y. Conversely, if the z_i generate, then for a prescribed vector (v', v_r) first solve

$$\sum_i c_i z_i = uv' - v_r w_j'.$$

Writing $w = \sum_i c_i w_i$, the vector

$$w + u^{-1}(v_r - w_r)w_j$$

equals (v', v_r). Thus surjectivity is equivalent to the finite disjunction, over j, of

$$a_j \text{ is a unit} \quad \text{and} \quad (z_i)_i \text{ is surjective onto } R^{r-1}.$$

Induction on r proves openness.

For a square matrix, put a unit entry of the last row in the last column and eliminate the other entries of that row. Surjectivity then reduces to surjectivity of the square upper-left block. By induction that block is invertible; the resulting block triangular matrix is invertible. This also includes the empty matrix.

Finally, apply induction to choose $r - 1$ of the z_i forming a basis. Take the corresponding w_i, together with w_j. The preceding formula solves every equation in these r columns. Hence their square matrix is invertible, and they form a basis. For $r = 0$ use the empty basis. □

We recall why open propositions and their families descend for this topology. First, $\neg\neg X_f$: otherwise evaluation would identify B_f with the one-element ring, contrary to its free basis containing 1 and $1 \neq 0$ in R. If $G_f \to P$ and P is open, then $\neg\neg P$, and stability of open propositions gives P. Thus every open proposition is a sheaf.

For an open predicate $U(x)$ on $X_f = \operatorname{Spec} B_f$, the proposition $\forall x\, U(x)$ is open. To see this, write U as a finite union of $D(g_i)$. This union covers precisely when the g_i generate 1 in B_f. Equivalently,

$$B_f^p \longrightarrow B_f, \qquad (b_i) \longmapsto \sum_i g_i b_i$$

is surjective. In the power basis this is surjectivity of a finite matrix over R, which is open by Lemma 5.1.

It follows that the type of open propositions, identified under logical equivalence, is a sheaf. Indeed, for a family of open propositions P_q indexed by $q \in G_f$, take the open proposition

$$P = \forall x \in X_f\ P_{|x|}.$$

For each $q \in G_f$, we have $P \leftrightarrow P_q$: one direction uses that G_f is a proposition, and the other eliminates $q : \|X_f\|$ into the proposition P_q. Uniqueness follows because implications between open propositions are sheaves.

Lemma 5.2. *If Q is a sheaf proposition, then "Q is open" is a sheaf proposition.*

Proof. Suppose $G_f \to (Q \text{ is open})$. Regard this as a family of open propositions, each with underlying proposition Q. The preceding result gives an open proposition P with $G_f \to (P \leftrightarrow Q)$. Both implications $P \to Q$ and $Q \to P$ are sheaves, so they hold without G_f. Hence $P \leftrightarrow Q$, and Q is open. □

Lemma 5.3. *Under* (2)*, any finite generating family in M merely contains a basis of length r.*

Proof. For a fixed tuple $v = (v_1, \dots, v_r)$ in M, the assertion that v is a basis is a sheaf proposition: it says that the fixed linear map $R^r \to M$ is an equivalence. By Lemma 5.2, its openness may be proved locally. There $M \cong R^r$, so it is invertibility of a square matrix, an open condition by Lemma 5.1.

Now let $w_1, \dots, w_s$ generate M. The proposition that some tuple $(w_{i_1}, \dots, w_{i_r})$ is a basis is a finite disjunction of open propositions, and hence is open and a sheaf. We may prove it locally using (2). For $M \cong R^r$ it follows from Lemma 5.1. □

6. Finitely many generators and the descent theorem

Proposition 6.1. *Continue to assume* (2). *Fix a monic f of positive degree. If $X_f \to \|\mathcal{B}_r(M)\|$, then M merely has finitely many generators.*

Proof. Put $X = X_f$ and $d = \deg f$. Zariski-local choice gives finitely many $g_i \in B_f$ generating 1, and, on each $D(g_i)$, a basis

$$v_{i1}(x), \dots, v_{ir}(x) \quad \text{of } M.$$

By Corollary 2.5 and Lemma 2.2, there are functions $w_{ij} : X \to M$ and exponents n_i such that

$$w_{ij}(x) = g_i(x)^{n_i} v_{ij}(x) \quad \text{on } D(g_i).$$

There are finitely many columns and charts, so these choices and a common exponent within each chart are legitimate finite choices. By Lemma 4.1, write

$$w_{ij}(x) = \sum_{k=0}^{d-1} x^k m_{ijk}, \qquad m_{ijk} \in M.$$

We claim that the finitely many elements m_{ijk} generate M.

Fix $m \in M$ and let $P(m)$ assert membership in their span. This is a sheaf proposition by Proposition 4.2. For each $x \in X$, some $g_i(x)$ is a unit, since the g_i generate 1 and R is local. Every $w_{ij}(x)$ is in the proposed span by its displayed expansion. Multiplying by $g_i(x)^{-n_i}$ shows that all the $v_{ij}(x)$ are in the span. They form a basis, so $P(m)$ holds. Thus $X \to P(m)$, and hence $G_f \to P(m)$. Since $P(m)$ is a sheaf, $P(m)$ holds globally. □

Proof of Theorem 1.1. Fix (2) and put $Q = \|\mathcal{B}_r(M)\|$. We show that Q is a sheaf. Given $G_f \to Q$, we have $X_f \to Q$. Proposition 6.1 supplies finitely many generators of M, and Lemma 5.3 supplies a basis of length r. Hence $(G_f \to Q) \to Q$ for every f, as required.

We can now eliminate (2) into the sheaf proposition Q, using the map which sends a basis to its mere existence. This yields $\|\mathcal{B}_r(M)\|$. □

The order of this proof matters. The assumption $\|\mathcal{B}_r(M)\|_T$ first proves the fraction condition, the coefficient isomorphism, descent of span membership, and openness of a fixed basis condition. Each of those targets is already known to be a sheaf proposition. Finitely many generators are then constructed under the separate hypothesis $X_f \to \|\mathcal{B}_r(M)\|$. Only after this construction do we prove that mere existence of a basis is itself a sheaf.

7. Bases with a retraction

Lemma 7.1. *Let $F : R^r \to R^m$ be fixed and linear. The proposition*

$$\exists H : R^m \to R^r \text{ linear} \qquad HF = 1$$

is a sheaf.

Proof. Write F_{ki} for the matrix of F. For each $j = 1, \ldots, r$, the jth row $(h_{j1}, \ldots, h_{jm})$ of a left inverse must satisfy

$$\sum_{k=1}^{m} h_{jk} F_{ki} = \delta_{ji} \qquad (1 \leq i \leq r).$$

Solvability is membership of the vector $(\delta_{ji})_i$ in the span of the fixed vectors $(F_{ki})_i$, for $1 \leq k \leq m$. It is a sheaf proposition by Proposition 4.2 applied to the already free module R^r. The conjunction of the r row conditions is a sheaf. Finite choice assembles solutions of the rows into H. □

Theorem 7.2. *Let E be a merely finite free R-module and let $S \subseteq E$ be a submodule such that membership in S is a sheaf proposition. If T-locally there are an isomorphism $b_0 : R^r \to S$ and a linear map $h_0 : E \to R^r$ with $h_0 \iota b_0 = 1$, where $\iota : S \hookrightarrow E$, then such a pair merely exists.*

Proof. We may choose a basis of E, since the conclusion asserts mere existence. The underlying type of E is a sheaf, and so is $S = \{e \in E : e \in S\}$. Forgetting h_0 gives local freeness of S; Theorem 1.1 gives a basis $b : R^r \to S$.

Fix this basis. Each local pair (b_0, h_0) gives a left inverse of ιb by the formula

$$h = b^{-1} b_0 h_0 : E \longrightarrow R^r.$$

Indeed, $h_0 \iota = b_0^{-1}$ because b_0 is an isomorphism, and hence $h \iota b = 1$. After identifying E with R^m, existence of this left inverse is a sheaf proposition by Lemma 7.1. Its local existence therefore gives mere existence of h globally. The pair (b, h) has the required properties. □

Corollary 7.3. *If an R-algebra A has a sheaf as its underlying type and is T-locally isomorphic to $M_N(R)$, then its underlying module merely has a basis of rank N^2. For a right ideal with sheaf membership, local existence of a rank-r basis with a linear retraction from A therefore implies mere existence of such a basis and retraction.*

Proof. The underlying module of $M_N(R)$ is free of rank N^2. Apply Theorem 1.1 to the underlying module of A, and then apply Theorem 7.2 to the right ideal as a submodule. □

References

[1] F. Cherubini, T. Coquand, M. Hutzler, *A foundation for synthetic algebraic geometry*, arXiv:2307.00073.